

Greg Hume (commons.wikimedia.org)

Trachemys scripta elegans (Wied-Neuwied, 1838)

Common name: Red-eared Slider

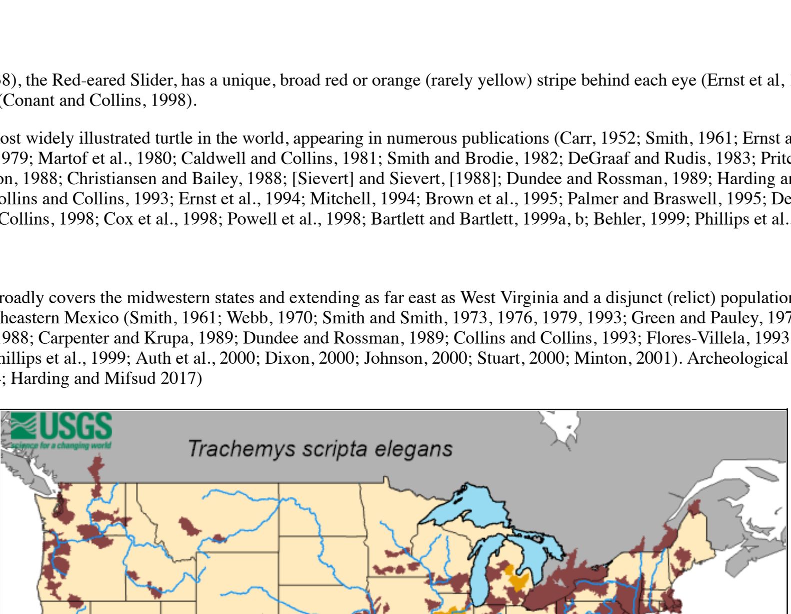
Taxonomy: available through ITIS

Identification: *Trachemys s. elegans* (Wied-Neuwied, 1838), the Red-eared Slider, has a unique, broad red or orange (rarely yellow) stripe behind each eye (Ernst et al, 1994; Conant and Collins, 1998). Younger individuals have numerous dark, eyelike spots on the yellow plastron (Conant and Collins, 1998).

Trachemys scripta, usually *T. s. elegans*, is probably the most widely illustrated turtle in the world, appearing in numerous publications (Carr, 1952; Smith, 1961; Ernst and Barbour, 1972, 1989; Mount, 1975; Behler and King, 1979; Pritchard, 1979; Sengoku, 1979; Smith and Smith, 1979; Martof et al., 1980; Caldwell and Collins, 1981; Smith and Brodie, 1982; DeGraaf and Rudis, 1983; Pritchard and Trebbau, 1984; [Mathui], 1985; Stebbins, 1985; Garrett and Barker, 1987; Green and Pauley, 1987; Alderton, 1988; Christiansen and Bailey, 1988; [Sievert] and Sievert, [1988]; Dundee and Rossman, 1989; Harding and Holman, 1990; Ashton and Ashton, 1991; Carmichael and Williams, 1991; Leviton et al., 1992; Collins, 1993; Collins and Collins, 1993; Ernst et al., 1994; Mitchell, 1994; Brown et al., 1995; Palmer and Braswell, 1995; Degenhardt et al., 1996; McKeown, 1996; Harding, 1997; Lamar, 1997; Branch, 1998; Campbell, 1998; Conant and Collins, 1998; Cox et al., 1998; Powell et al., 1998; Bartlett and Bartlett, 1999a, b; Behler, 1999; Phillips et al., 1999; Johnson, 2000; Minton, 2001).

Size: carapace length of 125-289 mm

Native Range: The Red-eared Slider's indigenous range broadly covers the midwestern states and extending as far east as West Virginia and a disjunct (relict) population in southern Ohio, as far west as eastern New Mexico, and as far south as south of the Rio Grande River into northeastern Mexico (Smith, 1961; Webb, 1970; Smith and Smith, 1973, 1976, 1979, 1993; Green and Pauley, 1978; Caldwell and Collins, 1981; Lohoefener and Altig, 1983; Garrett and Barker, 1987; Christiansen and Bailey, 1988; Carpenter and Krupa, 1989; Dundee and Rossman, 1989; Collins and Collins, 1993; Flores-Villela, 1993; Degenhardt et al., 1996; Conant and Collins, 1998; Ballinger and Lynch, 1999; Bartlett and Bartlett, 1999b; Phillips et al., 1999; Auth et al., 2000; Dixon, 2000; Johnson, 2000; Stuart, 2000; Minton, 2001). Archeological evidence from Native American middens shows that they were historically present near Saginaw, MI (Holman, 1994; Harding and Mifsud 2017)



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Trachemys scripta elegans* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AZ	1962	2022	16	Agua Fria; Big Sandy; Brawley Wash; Hassayampa; Lower Gila; Lower Gila-Painted Rock Reservoir; Lower Salt; Lower Santa Cruz; Lower Verde; Middle Gila; Rillito; San Francisco; Upper San Pedro; Upper Santa Cruz; Upper Verde; Willcox Playa
CA	1976	2024	32	Aliso-San Onofre; California; Central Coastal; Coyote; Los Angeles; Lower American; Lower Sacramento; Lower San Joaquin River; Mattole; Middle Kern-Upper Tehachapi-Grapevine; Mojave; Monterey Bay; Russian; Salton Sea; San Diego; San Francisco Bay; San Francisco Coastal South; San Gabriel; San Pablo Bay; Santa Ana; Santa Barbara Coastal; Santa Monica Bay; South Fork American; South Fork Kern; Suisun Bay; Tomales-Drake Bays; Upper Cache; Upper Coon-Upper Auburn; Upper Cosumnes; Upper Deer-Upper White; Upper Mokelumne; Upper Putah
CO	1984	2022	16	Big Thompson; Cache La Poudre; Clear; Colorado Headwaters-Plateau; Fountain; Lower White; Middle South Platte-Cherry Creek; Middle South Platte-Sterling; North Fork Republican; Roaring Fork; San Luis; St. Vrain; Two Butte; Uncompahgre; Upper Arkansas; Upper South Platte
CT	1980	2024	5	Housatonic; New England Region; Quinnipiac; Saugatuck; Thames
DE	2022	2022	1	Chincoteague
DC	2022	2022	1	Middle Potomac-Anacostia-Occoquan
FL	1958	2022	29	Apalachee Bay-St. Marks; Big Cypress Swamp; Caloosahatchee; Cape Canaveral; Crystal-Pithlachascotee; Daytona-St. Augustine; Econfina-Steinhatchee; Everglades; Florida Bay-Florida Keys; Florida Southeast Coast; Hillsborough; Kissimmee; Lower Ochlockonee; Lower St. Johns; Lower Suwannee; Manatee; Oklawaha; Peace; Pensacola Bay; Santa Fe; Sarasota Bay; South Atlantic-Gulf Region; Southern Florida; St. Marys; Tampa Bay; Upper St. Johns; Upper Suwannee; Vero Beach; Withlacoochee
GA	2012	2021	3	Upper Chattahoochee; Upper Ocmulgee; Upper Oconee
GU	2004	2020	1	Guam
HI	1996	2025	4	Hawaii; Kauai; Maui; Oahu
ID	2012	2016	3	Lower Boise; Palouse; Upper Snake-Rock
IL	2009	2009	1	Lower Wabash
IN	1931	2018	6	Driftwood; Lower Wabash; St. Joseph; Tippecanoe; Upper White; Wildcat
IA	1978	2013	3	East Nishnabotna; Lower Cedar; Lower Wapsipinicon
KS	2013	2013	1	Coon-Pickerel
ME	1987	1987	2	Lower Kennebec River; Presumpscot
MD	1991	2022	4	Gunpowder-Patapsco; Middle Potomac-Anacostia-Occoquan; Middle Potomac-Catoctin; Patuxent
MA	1975	2022	5	Ashuelot River-Connecticut River; Cape Cod; Charles; Concord River; Outlet Connecticut River
MI	1924	2024	7	Black-Macatawa; Detroit; Huron; Muskegon; Pere Marquette-White; Raisin; Upper Grand
MN	2014	2014	7	Des Moines Headwaters; Le Sueur; South Fork Crow; Twin Cities; Upper Mississippi-Black-Root; Watonwan; Zumbro
MT	2008	2017	2	Upper Missouri; Upper Yellowstone-Lake Basin
NE	1983	2015	6	Big Papillion-Mosquito; Blackbird-Soldier; Salt; South Fork Big Nemaha; Tarkio-Wolf; Upper Big Blue
NV	2016	2022	1	Las Vegas Wash
NJ	1976	2025	7	Crosswicks-Neshaminy; Hackensack-Passaic; Lower Delaware; Middle Delaware-Musconetcong; Mullica-Toms; Raritan; Sandy Hook-Staten Island
NM	1994	2000	3	Elephant Butte Reservoir; Rio Grande-Albuquerque; Tularosa Valley
NY	1953	2024	23	Bronx; Buffalo-Eighteenmile; Conewango; East Branch Delaware; Hackensack-Passaic; Hudson-Hoosic; Hudson-Wappinger; Irondequoit-Ninemile; Lower Genesee; Lower Hudson; Mettawee River; Middle Delaware-Mongaup-Brodhead; Middle Hudson; Mohawk; Niagara River; Northern Long Island; Oak Orchard-Twelvemile; Rondout; Sandy Hook-Staten Island; Schoharie; Seneca; Southern Long Island; Upper Susquehanna
NC	1915	2022	12	Albemarle; Deep; Haw; Lower Catawba; Lower Yadkin; Pigeon; Rocky; Upper Catawba; Upper Dan; Upper French Broad; Upper Neuse; Upper Yadkin
OH	1951	2015	2	Ashtabula-Chagrin; Walhonding
OR	1985	2024	20	Clackamas; Lost; Lower Columbia-Clatskanie; Lower Columbia-Sandy; Lower Willamette; Mckenzie; Middle Columbia-Lake Wallula; Middle Fork Willamette; Middle Rogue; Middle Willamette; Molalla-Pudding; Pacific Northwest Region; Siuslaw; South Santiam; South Umpqua; Tualatin; Umatilla; Umpqua; Upper Willamette; Wilson-Trusk-Nestuccu
PA	1948	2024	33	Bald Eagle; Brandywine-Christina; Chautauqua-Conneaut; Chester-Sassafras; Conemaugh; Connoquenessing; Crosswicks-Neshaminy; French; Kiskiminetas; Lackawaxen; Lake Erie; Lehigh; Lower Allegheny; Lower Delaware; Lower Juniata; Lower Monongahela; Lower Susquehanna; Lower Susquehanna-Penns; Lower Susquehanna-Swatara; Lower West Branch Susquehanna; Middle Allegheny-Redbank; Middle Allegheny-Tionesta; Middle Delaware-Mongaup-Brodhead; Middle Delaware-Musconetcong; Monocacy; Schuylkill; Shenango; Upper Delaware; Upper Juniata; Upper Ohio; Upper Susquehanna-Lackawanna; Upper Susquehanna-Tunkhannock; Youghiogheny
PR	2007	2018	4	Cibuco-Guajataca; Eastern Puerto Rico; Puerto Rican Islands; Southern Puerto Rico
RI	2014	2022	1	Narragansett
SC	1995	2025	8	Calibogue Sound-Wright River; Coastal Carolina; Congaree; Cooper; Enoree; Saluda; Seneca; South Carolina Coastal
TN	2014	2019	1	South Fork Holston
TX	1929	2001	5	Pedernales; Rio Grande-Fort Quitman; Salt Basin; Tierra Blanca; Toyah
UT	2020	2020	1	Spanish Fork
VA	1980	2022	8	Hampton Roads; Lower James; Lower Rappahannock; Lynnhaven-Poquoson; Middle Potomac-Anacostia-Occoquan; Pamunkey; Shenandoah; Upper Dan
WA	1995	2024	10	Duwamish; Lake Washington; Lower Columbia-Sandy; Lower Crab; Lower Yakima; Nisqually; Nooksack; Puget Sound; Puyallup; Snohomish
WV	2020	2020	1	Elk
WI	2005	2022	6	Coon-Yellow; Des Plaines; Lower Wisconsin; Middle Rock; Milwaukee; Upper Rock

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: Pond sliders are aquatic, omnivorous generalists, which rarely leave water except to bask (Ernst et al., 1994; Brown et al., 1995; Conant and Collins, 1998; Cox et al., 1998). Adult turtles tend to be more herbivorous than juveniles, but both will opportunistically eat aquatic invertebrates (especially insects and mollusks), fish, frog eggs and tadpoles, aquatic snakes, and a wide variety of aquatic plants and algae (Ernst et al., 1994; Brown et al., 1995). Although they prefer quiet waters, *T. elegans* is highly adaptable and can tolerate anything from brackish waters, to manmade canals, and city park ponds (Ernst et al., 1994; Cox et al., 1998; Salzberg, 2000). The Red-eared Slider may wander far from water, and rapidly colonize any newly available habitat (Cox et al., 1998; Minton, 2001). In California, this ability to rapidly colonize available habitat may be aided by an extensive system of manmade canals and irrigation ditches. Females lay 2-23 eggs per clutch in a cavity dug out of the soil, and as many as five clutches may be oviposited per year (Gibbons, 1990; Ernst et al., 1994).

Means of Introduction: Throughout its nonindigenous range *T. scripta* is introduced primarily through pet releases and escapes; a situation which has continued for several decades since the 1930s, reaching a peak during the Teenage Mutant Ninja Turtle television cartoon craze of the late 1980s-early 1990s (Ernst et al., 1994; Mitchell, 1994; Brown et al., 1995; Frank and McCoy, 1995; Lamar, 1997; Branch, 1998; Cox et al., 1998; Albright, 1999; Bartlett and Bartlett, 1999a; Ota, 1999; Williams, 1999; Beebee and Griffiths, 2000; Salzberg, 2000; Thomas and Hartnell, 2000; Minton, 2001; Porras, 2002). Additionally, most species of turtles are subject to intense commercial trade in the Asian food market (Williams, 1999); therefore, the transport and potential release of *T. scripta* for food and aquaculture in Asia and U.S. Pacific states with proportionally large Asian populations cannot be ruled out. Stebbins (1985) claims that pond sliders may have been introduced to Baja California, Mexico, for food; but his identity of this species as *T. scripta* is uncertain.

As recently as 2001, a Dutch animal welfare group attempted to ship unwanted pet *T. s. elegans* to Italy for eventual nonindigenous release into the wild (M. Hoogmoed and J. Perala as communicated to K. Dodd, personal communication 2001).

Status: In nearly all of its nonindigenous localities in Canada, New York, New Jersey, Pennsylvania, Michigan, Virginia, North Carolina, South Carolina, and perhaps Massachusetts, isolated populations of *T. s. elegans* are reproducing and thriving (Stein et al., 1980; Manchester, 1982; DeGraaf and Rudis, 1983; Harding and Holman, 1990; Cardoza et al., 1993; Ernst et al., 1994; Mitchell, 1994; D'Alessandro and Ernst, 1995; Palmer and Braswell, 1995; Platt and Snyder, 1996; Harding, 1997; Salzberg, 2000; Breisch et al., 2001; R. Burke, personal communication 1997; A. Hulse, personal communication 1997); although Klemens (1993) questions whether *T. s. elegans* eggs can hatch at colder latitudes in Connecticut. Klemens (1993) may be unaware of breeding populations found in Ontario, Canada, and Michigan; therefore; his doubts probably have no basis in fact. In the state of New York, *T. s. elegans* seem to be invasive, particularly on Long Island; however, it is unclear if related *T. s. scripta*, a race that prefers a warmer climate, can reproduce or even survive long at this latitude. Nonindigenous Yellow-bellied Sliders are established in Fairfax County, northern Virginia (Ernst et al., 1997).

The Red-eared Sliders found in Cumberland and Kennebec Counties, Maine, are surviving through the winter months, but it is not known if they are reproducing (Albright, 1999).

In Maryland, Red-eared Sliders are well established and invasive in a broad region throughout much of the northern and central portions of the state (Harris, 1975; Conant and Collins, 1998).

In Florida, most small populations of *T. s. elegans* are established in limited, small colonies (Moler, 1988; Ashton and Ashton, 1991; Bartlett, 1994; Dalrymple, 1994; Butterfield et al., 1997; Bartlett and Bartlett, 1999a; King, 2000; Townsend et al., 2002), except for those discovered in the Keys where there is yet no evidence of reproductive colonies (Butterfield et al., 1994; Duquesnel, 1996). A forthcoming assessment of the status of nonindigenous pond sliders in Florida will be made by Meshaka et al. (2003).

There is no evidence for an established population of nonindigenous *T. s. elegans* in Wisconsin (Cochran et al., 1987).

The current populations of *T. s. elegans* in Marion County, Indiana, are established and spreading, yet it is not clear if they represent an indigenous migration or nonindigenous introduction (Minton, 2001).

In Nebraska, there do not seem to be any current, established, nonindigenous *T. s. elegans*, and the only indigenous population occurs in Richardson County in the extreme southeast corner of the state (Ballinger and Lynch, 1999).

All Red-eared Sliders found in Colorado seem to be waifs from pet releases (Livo et al., 1998); therefore, this species is not listed by Hammerson (1999) as being part of Colorado's herpetofauna.

Nonindigenous populations of *T. s. elegans* in western Texas, New Mexico, and Arizona seem to be established (Hulse, 1980; Stebbins, 1985; Jones, 1988; Stuart, 1995a, 2000; Degenhardt et al., 1996; Howland, 1996; Dixon, 2000; M. Demlong, personal communication 1997), and the populations in the Rio Grande drainage in New Mexico are probably invasive. The *T. s. scripta* found in Arizona and the Rio Grande drainage of New Mexico do not yet represent established populations (Hulse, 1980; Stuart, 1995b, 2000; Degenhardt et al., 1996), but future surveys are required to establish this.

In the Pacific states of California, Oregon, and Washington, *T. s. elegans* is well established in numerous localities (Brown et al., 1995; Williams, 1999; Dvornich et al., 2001; D. Holland, personal communication 1997; L. Overtree, personal communication 1997; H. B. Schaffer, personal communication 2000; J. Abel, personal communication 2001). The records provided in this review may greatly underestimate the distribution of *T. scripta* in these states.

In Hawaii, Guam, and the Northern Mariana Islands, *T. s. elegans* is established but impacts are unclear (McCoid, 1993; Ernst et al., 1994; McCoid and Kleberg, 1995; McKeown, 1996).

Red-eared Sliders are well established on Guadeloupe, Lesser Antilles (Schwartz and Henderson, 1985, 1991; Censky and Kaiser, 1999).

In most cases, *T. s. elegans* seem to be established in its various non-U.S., worldwide localities mentioned above (Ernst et al., 1994; Lamar, 1997; Cox et al., 1998; Salzberg, 2000; Minton, 2001), except perhaps, New Zealand (Thomas and Hartnell, 2000). However, Beebee and Griffiths (2000) do not believe summers in the United Kingdom are long enough or warm enough for successful incubation of *T. s. elegans* eggs. In various parts of Southeast Asia, Red-eared Sliders are present (Cox et al., 1998).

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological	Economic	Human Health	Other

Trachemys scripta compete with indigenous species for food and basking sites (Frank and McCoy, 1995; Williams, 1999; Salzberg, 2000). In the U.S. state of Washington, they are a potential threat to Clemmys marmorata, the Pacific pond turtle (Williams, 1999), a declining species endemic to the Pacific states (Brown et al., 1995).

The spread of Red-eared Sliders in southern U.S. states, where the closely related Yellow-bellied Slider is indigenous, has the potential for *T. s. elegans* to compete with *T. s. scripta*. Additionally, interbreeding between the two races has occurred in various southern states (Ernst et al., 1994; Mitchell, 1994; Palmer and Braswell, 1995; Williams, 1999), which compromises the genetic integrity of indigenous Yellow-bellied Sliders. In New Mexico, T. s. elegans can cause a similar problem by hybridizing with Trachemys gaigae (Seidel et al., 1999; Stuart, 2000).

Aquaculture of Red-eared Sliders for the pet trade are often associated with the threat of salmonellosis in children who receive them as pets (Williams, 1999), but it could negatively impact indigenous wild turtles when released pets spread diseases and parasites into the environment (Stuart, 2000).

Remarks: The most current taxonomic reviews or summaries of pond sliders are by Seidel (1988, 2002), Seidel et al. (1999), and Iverson et al. (2000). Liner (1994) provides a list of regional vernacular names for *T. scripta*. The best summaries or literature reviews of the natural history of *T. scripta* are by Gibbons and Semlitsch (1991), Ernst et al. (1994), and various studies compiled by Gibbons (1990). Scientific and standard English names follow Crother (2008).

Pond sliders are commercially raised in incredibly large numbers for the pet trade throughout the southeastern U.S., especially Louisiana (Ernst et al., 1994; Williams, 1999; Salzberg, 2000). The sale of all turtles under the size of "4 inches" (about 100 mm) or viable eggs was banned in the United States, except for educational or research purposes, by the U.S. Food and Drug Administration in 1975 to prevent the spread of salmonellosis in children (Ernst et al., 1994; Level, 1997; Williams, 1999). Four inches or less is thought to be the size that a child can easily stuff into his mouth (Williams, 1999). The extensive use of antibiotics at turtle hatcheries in the U.S. has given rise to more antibiotic-resistant strains of *Salmonella* (Ernst et al., 1994). In many U.S. states *T. s. elegans* is the subject of specific regulation (Level, 1997). Nevertheless, undersized juvenile *T. scripta* are still commonly sold in pet stores in Florida, and as recently as 2002 they were sold by roadside venders at a busy intersection in Gainesville, Florida (Somma, personal observation). The European Union has banned the import of *T. s. elegans* because of the negative impact that released pets can have on native European pond turtles (*Emys orbicularis* species complex) (Williams, 1999).

References (click for full references)

Abel, J. 2001. Personal communication—Resident, Palo Alto, California.

Albright, J. 1999. Hypotheticals, accidentals, and other oddities. Pp. 197-200. In: M. L. Hunter, A. J. K. Calhoun, and M. McCollough (editors). Maine Amphibians and Reptiles. The University of Maine Press, Orono, Maine. 254 pp. + CD.

Alderton, D. 1988. Turtles & Tortoises of the World. Facts on File, Inc., New York. 191 pp.

Ashton, R. E., Jr., and P. S. Ashton. 1991. Handbook of Reptiles and Amphibians of Florida. Part Two. Lizards, Turtles & Crocodilians. Revised Second Edition. Windward Publishing, Inc., Miami. 191 pp.

Auth, D. L., H. M. Smith, B. C. Brown, and D. Lintz. 2000. A description of the Mexican amphibian and reptile collection of the Strecker Museum. Bulletin of the Chicago Herpetological Society 35(4):65-85.

Ballinger, R. E., and J. D. Lynch. 1999. Geographic distribution: *Trachemys scripta elegans* (red-eared slider). USA: Nebraska: Richardson Co. Herpetological Review 30(2):108-109.

Bartlett, R. D. 1994. Florida's alien herps. Reptile & Amphibian Magazine (27):56-73, 103-109.

Bartlett, R. D., and P. P. Bartlett. 1999a. A Field Guide to Florida Reptiles and Amphibians. Gulf Publishing Company, Houston. 280 pp.

Bartlett, R. D., and P. D. Bartlett. 1999b. A Field Guide to Texas Reptiles and Amphibians. Gulf Publishing Company, Houston. 331 pp.

Beebee, T. J. C., and R. A. Griffiths. 2000. Amphibians and Reptiles. A Natural History of the British Herpetofauna. HarperCollins Publishers, London. 270 pp.

Behler, J. L. 1999. National Audubon Society First Field Guide. Reptiles. Scholastic, Inc., New York. 160 pp.

Behler, J. L., and F. W. King. 1979. The Audubon Society Field Guide to North American Reptiles and Amphibians. Alfred A. Knopf, Inc., New York. 743 pp.

Beltz, E. 1997. HerPet-POURRI. Quite a range extension. Bulletin of the Chicago Herpetological Society 32(6):140.

Beltz, E. 2002. HerPET-POURRI. World traveled turtles. Bulletin of the Chicago Herpetological Society 37(6):112.

Branch, B. [=W. R.] 1998. Field Guide to Snakes and Other Reptiles of Southern Africa. Third Revised Edition. Ralph Curtis Books Publishing, Sanibel Island, Florida. 399 pp.

Breisch, A. R., J. W. Ozard, and 1425 Atlas Contributors. 2001. New York State Amphibian and Reptile Atlas Project [online]. Available on URL: http://www.dec.state.ny.us/website/dfwmr/wildlife/herp/turtles.html. New York State Department of Environmental Conservation, Division of Fish, Wildlife and Marine Resources, New York.

Brown, H. A., R. B. Bury, D. M. Darda, L. V. Diller, C. R. Peterson, and R. M. Storm. 1995. Reptiles of Washington and Oregon. Seattle Audubon Society, Seattle. 176 pp.

Burke, R. L. 1997. Personal communication—Zoologist/Professor, Department of Biology, Hofstra University, Hempstead, New York 11549.

Bury, R. B., and R. A. Luckenbach. 1976. Introduced amphibians and reptiles in California. Biological Conservation 1976(10):1-14.

Butterfield, B. P., W. E. Meshaka, Jr., and C. Guyer. 1997. Nonindigenous amphibians and reptiles. Pp. 123-138. In: D. Simberloff, D. C. Schmitz, and T. C. Brown (editors). Strangers in Paradise. Impact and Management of Nonindigenous Species in Florida. Island Press, Washington, D. C. 467 pp.

Butterfield, B. P., W. E. Meshaka, Jr., and J. B. Hauge. 1994. Two turtles new to the Florida Keys. Herpetological Review 25(2):81.

Caldwell, J. P., and J. T. Collins. 1981. Turtles in Kansas. AMS Publishing, Lawrence, Kansas. 67 pp.

Cardoza, J. E., G. S. Jones, T. W. French, and D. B. Halliwell. 1993. Exotic and Translocated Vertebrates of Massachusetts. Fauna of Massachusetts Series 6. Massachusetts Division of Fisheries and Wildlife, Westborough, Massachusetts. 106 pp.

Campbell, J. A. 1998. Amphibians and Reptiles of Northern Guatemala, the Yucatán, and Belize. University of Oklahoma Press, Norman, Oklahoma. 380 pp.

Carmichael, P., and W. Williams. 1991. Florida's Fabulous Reptiles and Amphibians. World Publications, Tampa. 120 pp.

Carpenter, C. C., and J. J. Krupa. 1989. Oklahoma Herpetology. An Annotated Bibliography. University of Oklahoma Press, Norman. 258 pp.

Carr, A. 1952. Handbook of Turtles. The Turtles of the United States, Canada, and Baja California. Comstock Publishing Associates, Ithaca and London. 542 pp.

Censky, E. J., and H. Kaiser. 1999. The Lesser Antillean fauna. Pp. 181-221. In: B. I. Crother (editor). Caribbean Amphibians and Reptiles. Academic Press, San Diego. 495 pp.

Christiansen, J. L., and R. M. Bailey. 1988. The lizards and turtles of Iowa. Iowa Department of Natural Resources Technical Series (3):1-19.

Cochran, P. A., J. B. Hodgson, and R. M. Korb. 1987. New distributional records for reptiles and amphibians in Brown County, Wisconsin. Herpetological Review 18(1):21-23.

Collins, J. T. 1993. Amphibians and Reptiles in Kansas. Third Edition, Revised. Museum of Natural History, The University of Kansas, Lawrence. 397 pp.

Collins, J. T., and S. L. Collins. 1993. Reptiles and Amphibians of the Cheyenne Bottoms. Hearth Publishing, Hillsboro, Kansas. 92 pp.

Conant, R., and J. T. Collins. 1998. A Field Guide to Reptiles & Amphibians. Eastern and Central North America. Third Edition, Expanded. Houghton Mifflin Company, Boston. 616 pp.

Cooper, J. E. 1959. The turtle *Pseudemys scripta* feral in Maryland. Herpetologica 15(1):44.

Cox, M. J., P. P. van Dijk, J. Nabhitabhata, and K. Thirakhupt. 1998. A Photographic Guide to Snakes and Other Reptiles of Peninsular Malaysia, Singapore and Thailand. New Holland Publishers (UK) Ltd, London. 144 pp.

Crother, B.I. (chair). Committee on Standard and English and Scientific Names. 2008. Scientific and standard English names of amphibians and reptiles of North America north of Mexico, with comments regarding confidence in our understanding. Society for the Study of Amphibians and Reptiles Herpetological Circular. No. 37. iii + 86p.

D'Alessandro, S. E., and C. H. Ernst. 1995. Additional geographic records for reptiles in Virginia. Herpetological Review 26(4):212-214.

Dalrymple, G. H. 1994. Non-indigenous amphibians and reptiles in Florida. Pp. 67-78. In: D. C. Schmitz and T. C. Brown (editors). An Assessment of Invasive Non-indigenous Species in Florida's Public Lands. Division of Environmental Resource Permitting, Florida Department of Environmental Protection, Technical Report No. TSS-94-100, Tallahassee. 303 pp.

Degenhardt, W. G., and J. L. Christiansen. 1974. Distribution and habitats of turtles in New Mexico. The Southwestern Naturalist 19(1):21-46.

Degenhardt, W. G., C. W. Painter, and A. H. Price. 1996. Amphibians and Reptiles of New Mexico. University of New Mexico Press, Albuquerque. 419 pp.

DeGraaf, R. M., and D. D. Rudis. 1983. Amphibians and Reptiles of New England. Habitats and Natural History. The University of Massachusetts Press, Amherst. 85 pp.

Demlong, M. 1997. Personal communication—Herpetologist, Phoenix Zoo, 455 N. Galvin Parkway, Phoenix, Arizona.

Dixon, J. R. 2000. Amphibians and Reptiles of Texas. Second Edition. Texas A & M University Press, College Station. 421 pp.

Dodd, K. 2001. Personal communication—Herpetologist, Florida Caribbean Science Center, Biological Resources Division, United States Geological Survey, 7920 NW 71st Street, Gainesville, Florida 32653.

Dundee, H. A., and D. A. Rossman. 1989. The Amphibians and Reptiles of Louisiana. Louisiana State University Press, Baton Rouge and London. 300 pp + unattached erratum.

Duquesnel, J. 1996. Scaly Visitors. Resource Management Notes (Florida Department of Environmental Protection, Tallahassee) 8(2):40.

Dvornich, K., and contributors. 2001. NatureMapping wildlife observations with the Washington Gap Analysis Project's predicted distributions. In: The NatureMapping Program in Washington [online]. Available on URL: http://www.fish.washington.edu/naturemapping/. University of Washington, Seattle.

Edgren, R. A., Jr. 1943. *Pseudemys scripta troostii* in Michigan. Copeia 1943(4):249.

Edgren, R. A., Jr. 1948. Some additional notes on Michigan *Pseudemys*. Natural History Miscellanea (Chicago) (22):1-2.

Ernst, C. H., and R. W. Barbour. 1972. Turtles of the United States and Canada. The University Press of Kentucky, Lexington. 347 pp.

Ernst, C. H., and R. W. Barbour. 1989. Turtles of the World. Smithsonian Institution Press, Washington, D.C. and London. 313 pp.

Ernst, C. H., S. C. Belfit, S. W. Sekscienski, and A. F. Laemmerzahl. 1997. The amphibians and reptiles of Ft. Belvoir and northern Virginia. Bulletin of the Maryland Herpetological Society 33(1):1-62.

Ernst, C. H., J. E. Lovich, and R. W. Barbour. 1994. Turtles of the United States and Canada. Smithsonian Institution Press, Washington and London. 578 pp.

Ernst, C. H., and J. E. Lovich. 2009. Turtles of the United States and Canada. JHU Press.

Fimrite, P. 2004. Turtles battle for Marin turf: Native western pond species is being driven out of its home in Marin water district lakes. [online] 2004(May 17). Available at URL http://SFGate.com

Flores-Villela, O. 1993. Herpetofauna Mexicana. Carnegie Museum of Natural History Special Publication (17):i-iv, 1-73.

Frank, J. H., and E. D. McCoy. 1995. Introduction to insect behavioral ecology: The good, the bad, and the beautiful: Non-indigenous species in Florida. Invasive adventive insects and other organisms in Florida. Florida Entomologist 78(1):1-15.

Garrett, J. M., and D. G. Barker. 1987. A Field Guide to Reptiles and Amphibians of Texas. Texas Monthly Press, Austin. 225 pp.

Gibbons, J. W. (editor). 1990. Life History and Ecology of the Slider Turtle. Smithsonian Institution Press, Washington, D.C. and London. 368 pp.

Gibbons, J. W., and R. D. Semlitsch. 1991. Guide to the Reptiles and Amphibians of the Savannah River Site. The University of Georgia Press, Athens, Georgia, and London. 131 pp.

Gordon, H., and J. A. Fowler. 1961. A new locality record for *Pseudemys scripta elegans* in Michigan. Copeia 1961(3):350.

Green, N. B., and T. K. Pauley. 1987. Amphibians and Reptiles in West Virginia. University of Pittsburgh Press, Pittsburgh. 241 pp.

Hammerson, G. A. 1999. Amphibians and Reptiles in Colorado. Second Edition. University Press of Colorado, Niwot, Colorado. 484 pp.

Harding, J. H., and J. A. Holman. 1990. Michigan Turtles and Lizards. Department of Outreach Communications, Michigan State University, East Lansing. 94 pp.

Harding, J. H. 1997. Amphibians and Reptiles of the Great Lakes Region. The University of Michigan Press, Ann Arbor. 378 pp.

Harding, J.H. and D.A. Mifsud. 2017. Amphibians and reptiles of the Great Lakes region. Revised edition. The University of Michigan Press, Ann Arbor, MI.

Harris, H. S., Jr. 1975. Distributional survey (Amphibia/Reptilia): Maryland and the District of Columbia. Bulletin of the Maryland Herpetological Society 11(3):73-167.

Holland, D. C. 1997. Personal communication—Herpetologist, Fallbrook, California.

Holman, A.J. 1994. Status of the Red-Eared Slider turtle Trachemys scripta elegans (Wied) in Michigan: a preliminary report. Michigan Academician XXVI: 471-477.

Howland, J. M. 1996. Herps of Arizona. The Desert Monitor (Phoenix) 27(1):12-17.

Hudson, B., and T. J. Thornton. 1994. Reptiles & Amphibians in New Zealand: Handbook for Species Identification. Print Media Specialists, Auckland. 50 pp

Hudson, G. E. 1942. The amphibians and reptiles of Nebraska. Nebraska Conservation Bulletin (24):[i-ii], 1-145. (Reprinted 1972.)

Hulse, A. C. 1980. Notes on the occurrence of introduced turtles in Arizona. Herpetological Review 11(1):16-17.

Hulse, A. S. 1997. Personal communication—Herpetologist/Professor, Indiana University of Pennsylvania, Indiana, Pennsylvania.

Hutchison, A. M. 1992. A reproducing population of *Trachemys scripta elegans* in southern Pinellas County, Florida. Herpetological Review 23(3):74-75.

Iverson, J. B. 1992. A Revised Checklist with Distribution Maps of the Turtles of the World. John B. Iverson, Richmond, Indiana. 363 pp.

Iverson, J. [B.], P. [A.] Meylan, and M. [E.] Seidel. 2000. Testudines—turtles. Pp. 75-82. In: B. I. Crother (chair), and Committee on Standard English and Scientific Names (editors). Scientific and standard English names of amphibians and reptiles of North America north of Mexico, with comments regarding confidence in our understanding. Society for the Study of Amphibians and Reptiles Herpetological Circular (29):i-iii, 1-82.

Johnson, T. R. 2000. The Amphibians and Reptiles of Missouri. Revised and Expanded Second Edition. Missouri Department of Conservation, Conservation Commission of the State of Missouri, Jefferson City. 400 pp.

Jones, B. K. 1988. Distribution and habitat associations of herpetofauna in Arizona: Comparisons by habitat type. Pp. 109-128. In: R. C. Szaro, K. E. Severson, and D. R. Patton (coordinators). Management of Amphibians, Reptiles, and Small Mammals in North America. Proceedings of the Symposium. July 19-21, 1988, Flagstaff, Arizona. United States Department of Agriculture Forest Service, General Technical Report RM-166, Fort Collins. 458 pp.

King, F. W. 2000. Florida Museum of Natural History's Checklist of Florida Amphibians and Reptiles [online]. Available on URL: http://www.flmnh.ufl.edu/herps/FL-GUIDE/Flaherps.htm. Florida Museum of Natural History, University of Florida, Gainesville.

King, F. W., and R. L. Burke (editors). 1989. Crocodilian, Tuatara, and Turtle Species of the World. A Taxonomic and Geographic Reference. The Association of Systematics Collections, Washington, DC. 216 pp.

Klemens, M. W. 1993. Amphibians and reptiles of Connecticut and adjacent regions. State Geological and Natural History Survey of Connecticut Bulletin (112):i-xii, 1-318.

Lamar, W. W. 1997. The World's Most Spectacular Reptiles & Amphibians. World Publications, Tampa. 210 pp.

Level, J. P. 1997. A Field Guide to Reptiles and the Law. Second Revised Edition. Serpent's Tale Natural History Book Distributors, Lanesboro, Minnesota. 270 pp.

Leviton, A. E., S. C. Anderson, K. Adler, and S. A. Minton, [Jr.] 1992. Handbook to Middle East Amphibians and Reptiles. Contributions to Herpetology 8. Society for the Study of Amphibians and Reptiles, Oxford, Ohio. 252 pp.

Liner, E. A. 1994. Scientific and common names for the amphibians and reptiles of Mexico in English and Spanish. Nombres científicos y comunes en Ingles y Españole de los anfibios y los reptiles de México. Society for the Study of Amphibians and Reptiles Herpetological Circular (23):i-vi, 1-113.

Livo, L. J., G. A. Hammerson, and H. M. Smith. 1998. Summary of amphibians and reptiles introduced into Colorado. Northwestern Naturalist 79(1):1-11.

Lohoefener, R., and R. Altig. 1983. Mississippi herpetology. Mississippi State University Research Center Bulletin (1):i-vi, 1-66.

[Lue Guangyang, Du Mingzhang, and Xiang Gaoshi. 1999. Taiwan Liang Qi Pa Xing Doang Wu Tu Jian. Chu Ban. Zhonghua Min Guo Zi Ran Sheng Tai Bao Yu Xie Hui, Da Zi Ran Za Zhi She, Taibei. (A Pictorial Guide to the Amphibians and Reptiles of Taiwan. Chu Ban. ROC & Great Nature Press, Taipei.) In Chinese]. 343 pp.

Lynch, J. D. 1985. Annotated checklist of the amphibians and reptiles of Nebraska. Transactions of the Nebraska Academy of Sciences 23:33-57.

Manchester, D., Sr. 1982. Red-eared sliders in Pennsylvania. Testudo 2(1):27-30.

Martof, B. S. 1956. Amphibians and Reptiles of Georgia. University of Georgia Press, Athens, Georgia. 94 pp.

Martof, B. S., W. M. Palmer, J. R. Bailey, and J. R. Harrison III. 1980. Amphibians and Reptiles of the Carolinas and Virginia. The University of North Carolina Press, Chapel Hill. 264 pp.

[Mathui, K.] 1985. [Amphibians and Reptiles of Japan. Shôgakûkau, Tokyo.] [In Japanese.] 160 pp.

McCann, J. A., L. N. Arkin, and J. D. Williams. 1996. Nonindigenous Aquatic and Selected Terrestrial Species in Florida. Status, Pathways, Dates of Introduction, Range Distributions, and Significant Ecological and Economic Effects. Florida Caribbean Science Center, U. S. Geological Survey, Gainesville. 301 pp.

McCoid, M. J. 1993. The "new" Herpetofauna of Guam, Mariana Islands. Herpetological Review 24(1):16-17.

McCoid, M. J., and C. Kleberg. 1995. Non-native reptiles and amphibians. Pp. 433-437. In: E. T. LaRoe, G. S. Farris, C. E. Puckett, P. D. Doran, and M. J. Mac (editors). Our Living Resources: A Report to the Nation on the Distribution, Abundance, and Health of U. S. Ecosystems. U.S. Department of the Interior, National Biological Service, Washington, D. C. 530 pp.

McKeown, S. 1996. A Field Guide to Reptiles and Amphibians in the Hawaiian Islands. Diamond Head Publishing, Inc., Los Osos, California. 172 pp.

Meshaka, W. E., Jr., B. P. Butterfield, and J. B. Hauge. 2003 (in press). The Exotic Amphibians and Reptiles of Florida. Krieger Publishing Company, Malabar, Florida.

Minton, S. A., Jr. 2001. Amphibians & Reptiles of Indiana. Revised 2nd Edition. Indiana Academy of Science, Indianapolis. 404 pp.

Mitchell, J. C. 1994. The Reptiles of Virginia. Smithsonian Institution Press, Washington and London. 352 pp.

Moler, P. 1988. A Checklist of Florida's Amphibians and Reptiles. Nongame Wildlife Program, Florida Game and Fresh Water Fish Commission, Tallahassee. 18 pp.

Mount, R. H. 1975. The Reptiles and Amphibians of Alabama. Auburn University Agricultural Experiment Station, Auburn. 347 pp.

Murphy, J. C. 1997. Amphibians and Reptiles of Trinidad and Tobago. Krieger Publishing Company, Malabar, Florida. 245 pp.

Ota, H. 1999. Introduced amphibians and reptiles of the Ryukyu Archipelago, Japan. Pp. 439-452. In: G. H. Rodda, Y. Sawai, D. Chiszar, and H. Tanaka (editors). Problem Snake Management: The Habu and the Brown Treesnake. Cornell University Press, Ithaca. 534 pp.

Overtree, L. 1997. Personal communication—Biologist, South Fork Kern River Valley research, California.

Palmer, W. M., and A. L. Braswell.1995. Reptiles of North Carolina. The University of North Carolina Press for North Carolina State Museum of Natural Sciences, Chapel Hill and London. 412 pp.

Phillips, C. A., R. A. Brandon, and E. O. Moll. 1999. Field Guide to Amphibians and Reptiles of Illinois. Illinois Natural History Survey, Champaign, Illinois. 282 pp.

Platt, S. G., and W. E. Snyder. 1996. Geographic distribution: *Trachemys scripta elegans* (red-eared slider). USA: South Carolina: Pickens Co. Herpetological Review 27(3):151.

Porras, L. W. 2002. Meet the locals: Your guide to backyard herping. Part four: The East. Reptiles 10(9):92-99.

Powell, R., J. T. Collins, and E. D. Hooper, Jr. 1998. A Key to Amphibians & Reptiles of the Continental United States and Canada. University Press of Kansas, Lawrence. 131 pp.

Powell, R., R. Conant, and J.T. Collins. 2016. Peterson field guide to reptiles and amphibians of eastern and central North America. Houghton Mifflin Harcourt.

Pritchard, P. C. H. 1979. Encyclopedia of Turtles. T.F.H. Publications, Inc., Neptune, New Jersey. 895 pp.

Pritchard, P. C. H., and P. Trebbau. 1984. The Turtles of Venezuela. Contributions to Herpetology 2. Society for the Study of Amphibians and Reptiles, Ithaca. 403 pp., 47 plates, 16 maps.

Salzberg, A. 2000. The cage papers. The Norway rat of the turtle world. Reptile & Amphibian Hobbyist 5(8):84.

Schaffer, H. B. 2000. Personal communication—Herpetologist, Section of Evolution and Ecology, University of California, Davis, California 95616.

Schwartz, A., and R. W. Henderson. 1985. A Guide to the Identification of the Amphibians and reptiles of the West Indies Exclusive of Hispaniola. Milwaukee Public Museum, Milwaukee. 165 pp.

Schwartz, A., and R. W. Henderson. 1991. Amphibians and Reptiles of the West Indies: Descriptions, Distributions, and Natural History. University of Florida Press, Gainesville. 720 pp.

Schwartz, A., and R. Thomas. 1975. A check-list of West Indian amphibians and reptiles. Carnegie Museum of Natural History Special Publication (1):1-216.

Seidel, M. E. 1988. Revision of the West Indian emydid turtles (Testudines). American Museum Novitates (2918):1-41.

Seidel, M. E. 2002. Taxonomic observations on extant species and subspecies of slider turtles, genus *Trachemys*. Journal of Herpetology 36(2):285-292.

Seidel, M. E., J. N. Stuart, and W. G. Degenhardt. 1999. Variation and species status of slider turtles (Emydidae: *Trachemys*) in the southwestern United States and adjacent Mexico. Herpetologica 55(4):470-487.

Sengoku, S. 1979. [Amphibians and Reptiles. Ienohikari (House of Light) Corporation, Tokyo. In Japanese.] 206 pp.

[Sievert], G., and L. Sievert. [1988]. A Field Guide to Reptiles of Oklahoma. Oklahoma Department of Wildlife Conservation, Oklahoma City. 96 pp.

Smith, H. M., and E. D. Brodie, Jr. 1982. A Guide to Field Identification. Reptiles of North America. Golden Press, New York. 240 pp.

Smith, H. M., and A. J. Kohler. 1978. A survey of herpetological introductions in the United States and Canada. Transactions of the Kansas Academy of Science 1977 80(1-2):1-24.

Smith, H. M., and R. B. Smith. 1973. Synopsis of the Herpetofauna of Mexico. Volume II. Analysis of the Literature Exclusive of the Mexican Axolotl. John Johnson Natural History Books, North Bennington, Vermont. 367 pp.

Smith, H. M., and R. B. Smith. 1976. Synopsis of the Herpetofauna of Mexico. Volume III. Source Analysis and Index for Mexican Reptiles. John Johnson, North Bennington, Vermont. 23 pp., Am-T, App-102, Cor-4.

Smith, H. M., and R. B. Smith. 1979. Synopsis of the Herpetofauna of Mexico. Volume VI. Guide to Mexican Turtles. Bibliographic Addendum III. 1044 pp.

Smith, H. M., and R. B. Smith. 1993. Synopsis of the Herpetofauna of Mexico. Volume VII. Bibliographic Addendum IV and Index, Bibliographic Addenda II-IV, 1979-1991. University Press of Colorado, Niwot, Colorado. 1082 pp.

Smith, P. A. 1961. The amphibians and reptiles of Illinois. Illinois Natural History Survey Bulletin 28(1):[i-v], 1-298. (Reprinted 1986.)

Solis, F. 2001. Personal communication—Biologist, Fish and Wildlife Service, 3600 E. Paisano, Building A, Room 142, El Paso, Texas 79905.

Spinks, P. Q., G. B. Pauly, J. J. Crayon and H. B. Shaffer. 2003. Survival of the western pond turtle (Emys marmorata) in an urban California environment. Biological Conservation 113(2): 257-267.

Stebbins, R. C. 1972. California Amphibians and Reptiles. University of California Press, Berkeley, Los Angeles, and London. 152 pp.

Stebbins, R. C. 1985. A Field Guide to Western Reptiles and Amphibians. Second Edition. Houghton Mifflin Company, Boston. 336 pp.

Stein, R. J., W. K. Eames, and D. C. Parris. 1980. Geographic distribution: *Chrysemys scripta elegans* (red-eared slider). USA: New Jersey: Mercer Co. Herpetological Review 11(4):115.

Stevenson, H. S. 1976. Vertebrates of Florida. Identification and Distribution. University Presses of Florida, Gainesville. 607 pp.

Stuart, J. N. 1995a. Notes on the aquatic turtles of the Rio Grande drainage, New Mexico. Bulletin of the Maryland Herpetological Society 31(3):147-157.

Stuart, J. N. 1995b. Geographic distribution: *Trachemys scripta scripta* (yellowbelly slider). USA: New Mexico: Socorro Co. Herpetological Review 26(2):107.

Stuart, J. N. 2000. Additional notes on native and non-native turtles of the Rio Grande Drainage Basin, New Mexico. Bulletin of the Chicago Herpetological Society 35(10):229-235.

Thomas, M., and P. Hartnell. 2000. An occurrence of a red-eared turtle (*Trachemys scripta elegans*) in the Waikato River at Hamilton, New Zealand. Herpetofauna (Sydney) 30(2):15-17.

Townsend, J. H., K. L. Krysko, A. T. Reppas, and C. M. Sheehy III. 2002. Noteworthy new records for introduced reptiles and amphibians from Florida, USA. Herpetological Review 33(1):75.

Webb, R. G. 1970. Reptiles of Oklahoma. University of Oklahoma Press, Norman. 370 pp.

Williams, T. 1999. The terrible turtle trade. Audubon 101(2):44, 46-48, 50-51.

Other Resources:
Turtles battle for Marin turf: Native western pond species being driven out

Trachemys scripta elegans (Global Invasive Species Database)

NOBANIS: Northern European and Baltic Network on Invasive Alien Species

Author: Somma, L.A., Foster, A., Fuller, P. and C. Cameron

Revision Date: 4/3/2024

Citation Information:
Somma, L.A., Foster, A., Fuller, P. and C. Cameron, 2025, *Trachemys scripta elegans* (Wied-Neuwied, 1838): U.S. Geological Survey, Nonindigenous Aquatic Species Database, Gainesville, FL, https://nas.er.usgs.gov/queries/FactSheet.aspx?SpeciesID=1261, Revision Date: 4/3/2024, Access Date: 12/3/2025

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